

CHAIN SURVEYING - SUMMARY NOTES

1. CHAIN SURVEYING

Definition: Chain surveying is the simplest and most common method of surveying in which horizontal distances are measured directly on the ground using a chain or tape.

Principle: Based on the principle of triangulation - the area is divided into a series of triangles.

Purpose: Used for measuring small, fairly level areas.

Characteristics:

- Suitable for small areas with simple details.
- Simple, quick, and inexpensive.
- Only linear measurements are taken.

2. TYPES OF CHAINS

Type of Chain | Length | Use

Gunter's Chain | 66 ft (100 links) | Used in land measurement.

Revenue Chain | 33 ft (16 links) | Used for cadastral surveys.

Engineer's Chain | 100 ft (100 links) | Used in engineering surveys.

Metric Chain | 20 m (100 links) or 30 m (150 links) | Standard in India; easy to use and repair.

Advantages: Simple, easy to handle, durable.

Disadvantages: Heavy and subject to sagging or elongation.

3. INSTRUMENTS USED FOR CHAINING

- Pegs: Wooden or iron stakes for marking survey stations.
- Arrows: Steel pins for marking chain lengths.
- Ranging Rods: Used for aligning survey lines.
- Offset Rods: Used for short offsets perpendicular to chain line.
- Plumb Bob: Used for transferring points vertically.
- Line Ranger: Used to locate intermediate points accurately.

4. RANGING OPERATION

Definition: Ranging is the process of aligning intermediate points on a straight survey line between two end stations.

Types of Ranging:

- 1. Direct Ranging - When both end stations are visible. Uses visual alignment and hand signals.
- 2. Indirect Ranging - When end stations are not visible due to obstruction. Points are fixed by successive alignment.

5. OFFSETS

Definition: The lateral distance measured from the chain line to locate a boundary, feature, or detail.

Purpose: To locate the details near the survey line for plotting.

Types of Offsets:

- 1. Perpendicular Offset - Drawn at 90 degrees to chain line using cross-staff or optical square.
- 2. Oblique Offset - Drawn at an angle other than 90 degrees.

Limiting Length of Offset: Should be short to minimize plotting error depending on accuracy and scale.

QUICK REVISION TABLE

Topic | Key Points

Chain Surveying	Simple method using chains & tapes for horizontal distance.
Types of Chains	Gunter's, Revenue, Engineer's, Metric.
Instruments Used	Pegs, Arrows, Ranging Rods, Offset Rods, Plumb Bob, Line Ranger.
Ranging	Aligning points (Direct & Indirect).
Offsets	Perpendicular & Oblique offsets for locating features.